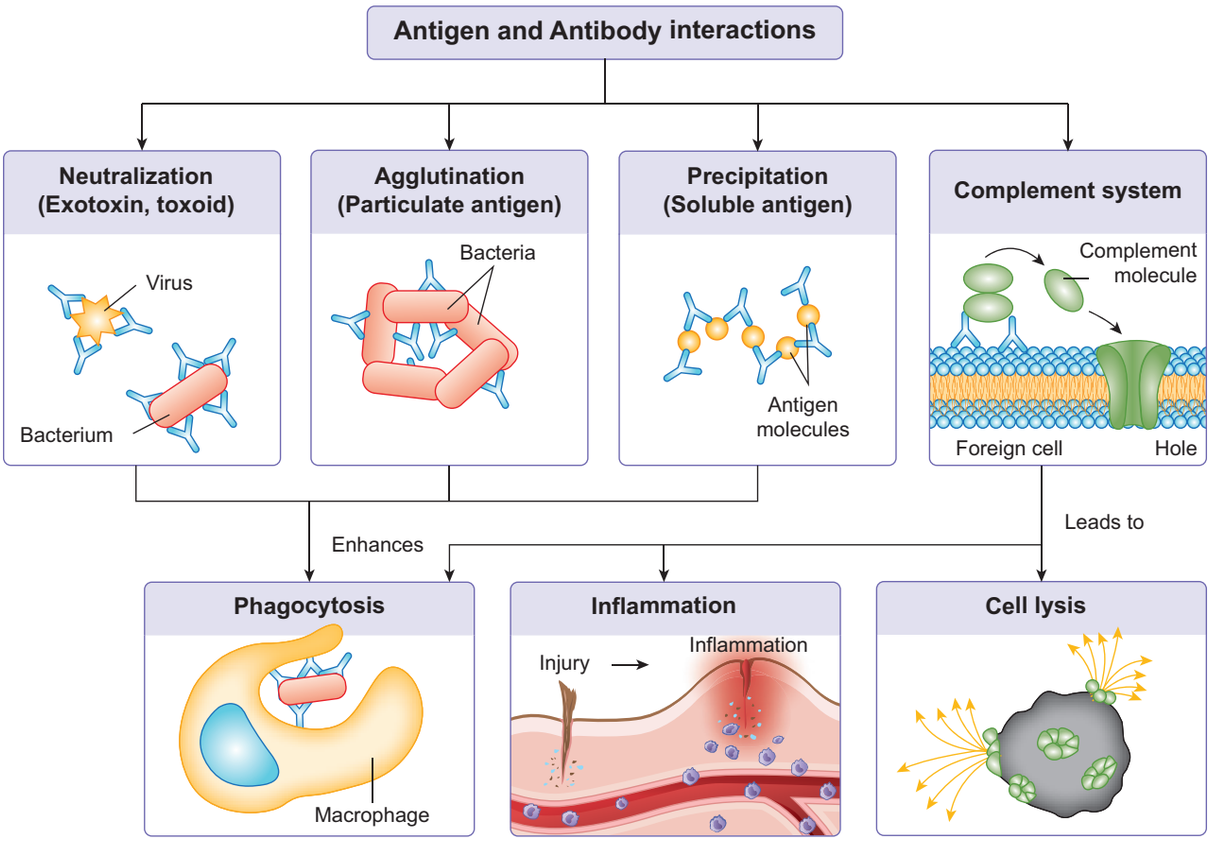


Fig. 8.8 Antigen and antibody reaction

a substance called an opsonin, marking the pathogen out for destruction by the immune system. This results in a much more **efficient phagocytosis**.

The **neutralization** reactions are the reactions of antigen-antibody that involve the **elimination of harmful effects** of bacterial exotoxins or a virus by specific antibodies. These neutralizing substances i.e., antibodies are known as **antitoxins**. This specific antibody is produced by a host cell in response to a bacterial exotoxin or corresponding toxoid (inactivated toxin).

8. 9 Vaccines

A vaccine **is a biological preparation that provides active acquired immunity** to a particular disease and resembles a

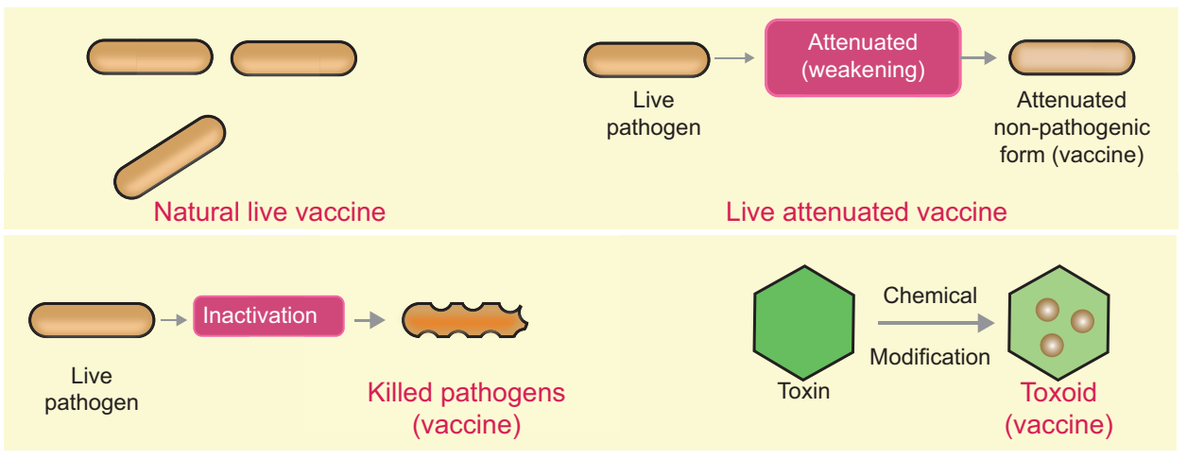


Fig. 8.9 Types of vaccines

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